



23PCPE402 SENSORS AND ACTUATORS

Project Based Learning (PBL)

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“AUTOMATIC TOUCHLESS DOOR CONTROL SYSTEM ”

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1.Introduction to project

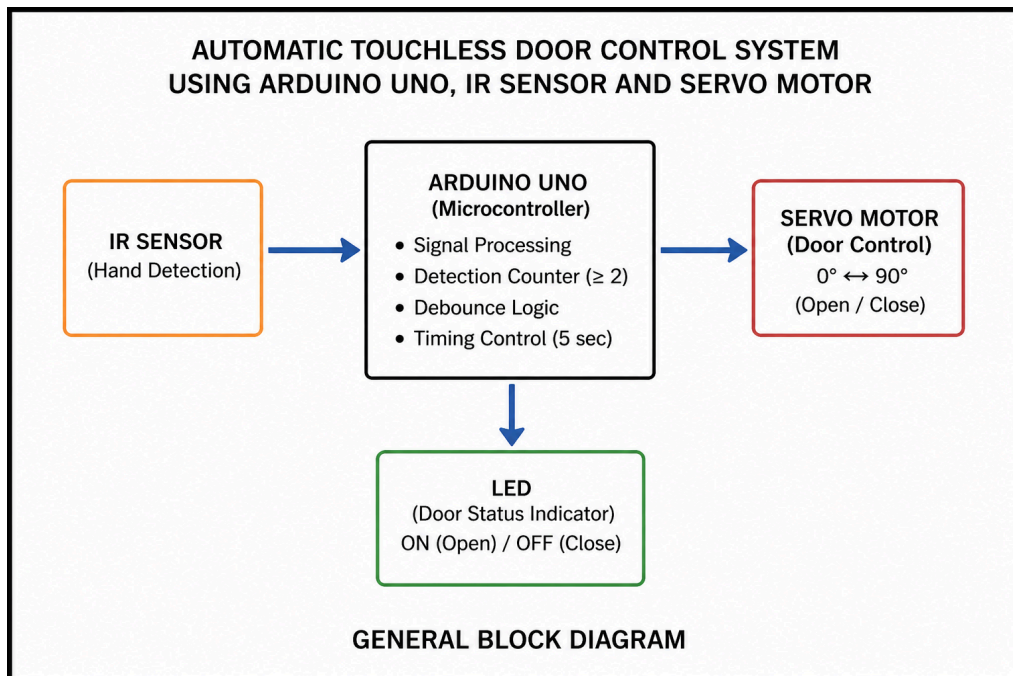
The Automatic Touchless Door Control System is a simple and efficient embedded system designed to open and close a door without physical contact. The system uses an IR proximity sensor to detect hand movement and an Arduino UNO microcontroller to process the input signal. Based on this detection, a servo motor is used to control the opening and closing of the door.

In today's world, touchless systems are highly important in places like hospitals, malls, and offices to maintain hygiene and convenience. This project demonstrates a low-cost and effective solution for automated access control using sensor–actuator integration.

2. Aim of PBL

To design and implement a touchless automatic door system using an **Arduino UNO, IR sensor and servo motor**, which detects hand movement and operates a door without physical contact.

3. General Block Diagram



4. Explanation of System

The system works on a simple sensor–actuator mechanism. When a user moves their

hand near the IR sensor, it detects the motion and sends a signal to the Arduino UNO. The Arduino processes this signal and checks if the detection occurs twice to avoid false triggering.

Once the condition is satisfied, the Arduino sends a signal to the servo motor to rotate from 0° to 90° , which opens the door. At the same time, an LED glows to indicate the door is open. After a delay of 5 seconds, the servo motor returns to its original position (0°), closing the door, and the LED turns off.

5. Selection of Components

1. Arduino UNO

- *Acts as the main controller*
- *Easy to program and widely used*
- *Provides digital and PWM pins*

2. IR Sensor

- *Detects hand movement using infrared rays*
- *Provides digital HIGH/LOW output*
- *Low cost and reliable for short-range detection*

3. Servo Motor (SG90)

- *Provides precise angle control (0° – 90°)*
- *Used to open/close the door*
- *Lightweight and easy to interface*

4. LED

- *Used as a status indicator*
- *Shows when the door is open*

5. Breadboard & Wires

- *Used for connections and prototyping*
-

6. Interfacing Sensors and Actuators with Microcontrollers

Interfacing IR Sensor with Arduino

- *OUTPUT pin $\rightarrow$ Arduino Pin 2*

- $VCC \rightarrow 5V$
- $GND \rightarrow GND$

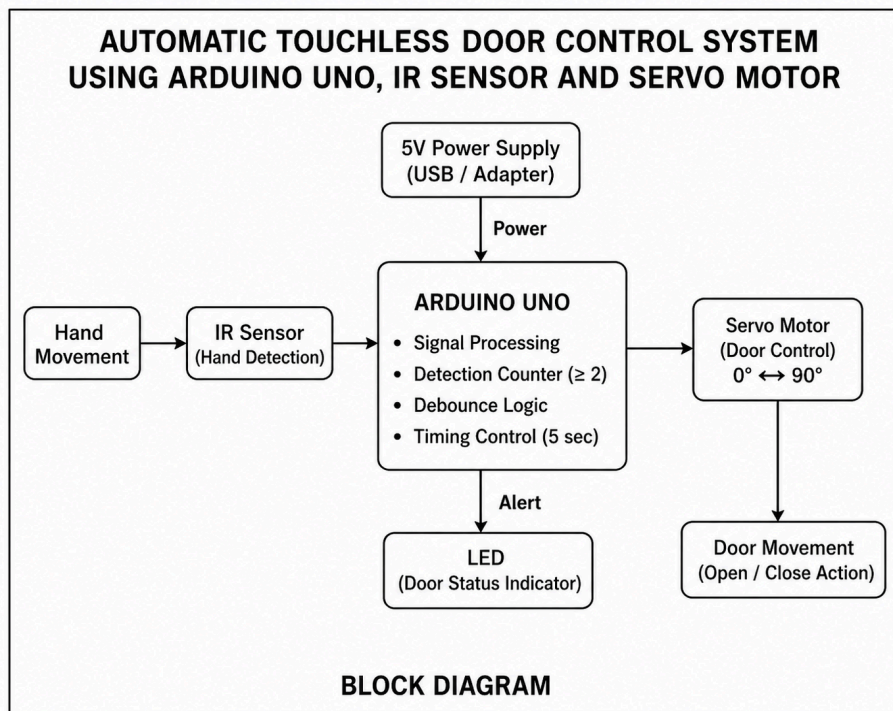
Interfacing Servo Motor

- $Signal \rightarrow Pin\ 6$
- $VCC \rightarrow 5V$
- $GND \rightarrow GND$

Interfacing LED

- $Anode \rightarrow Pin\ 7\ via\ resistor$
- $Cathode \rightarrow GND$

Block Diagram :



7. Explanation

(Imp- Detail Pin/Ports no. of sensors and microcontroller should be mentioned)

The IR sensor continuously monitors for hand movement. When a hand is detected, it sends a LOW signal to Arduino Pin 2. Each detection increases a counter.

When the counter reaches 2 (two detections), Arduino sends a PWM signal through Pin 6 to the servo motor. The servo rotates to 90° , opening the door.

Simultaneously:

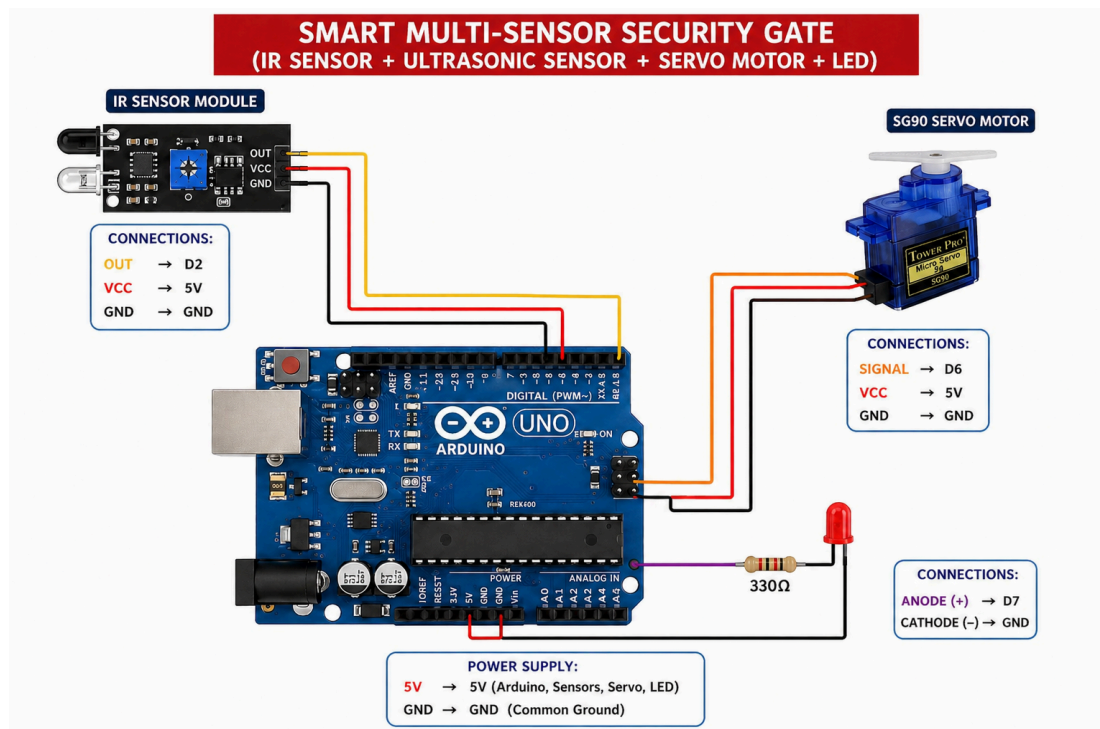
- LED on Pin 7 turns ON

After 5 seconds:

- Servo returns to 0°
- LED turns OFF
- Counter resets

This ensures reliable and intentional triggering.

8. Circuit Diagram



9. Explanation

The circuit consists of an Arduino UNO connected to an IR sensor, a servo motor, and an LED.

- The IR sensor detects hand movement and sends signals to Arduino.
- Arduino processes the signal and controls the servo motor.
- The servo motor rotates to open and close the door.
- The LED indicates the door status.

All components share a common ground and are powered using the Arduino 5V supply.

1. Working

The system operates in a sequence of sensing, processing, and actuation to achieve touchless door control.

1. Detection of Hand Movement

The IR sensor continuously emits infrared rays and monitors their reflection. When a hand is placed near the sensor, the reflected IR signal changes, and the sensor outputs a digital LOW signal. This indicates that an object (hand) is detected within its range.

2. Counting Mechanism in Arduino

Each time the IR sensor detects a hand, the Arduino reads the signal and increments a counter variable. A small delay (debounce delay) is used to ensure that a single hand movement is not counted multiple times due to noise or rapid fluctuations.

3. Two-Detection Condition for Activation

The system is designed to open the door only after detecting two consecutive hand movements. This condition helps prevent accidental triggering caused by random disturbances or environmental noise. When the counter reaches a value of 2, the Arduino confirms a valid user action.

4. Door Opening using Servo Motor

Once the condition is satisfied, the Arduino sends a control signal to the servo motor. The servo rotates from 0° (closed position) to 90° (open position), which mechanically opens the door. At the same time, the LED is turned ON to provide a visual indication that the door is open.

5. Time Delay for Open State

After opening, the door remains in the open position for a fixed duration of 5 seconds. This delay allows sufficient time for a person to pass through the door comfortably.

6. Automatic Door Closing

After the delay, the Arduino commands the servo motor to rotate back to 0° , returning the door to the closed position. Simultaneously, the LED is turned OFF, indicating that the door is closed.

7. System Reset for Next Cycle

Once the door is closed, the counter is reset to zero, and the system returns to its initial state. It is now ready to detect the next set of hand movements and repeat the process.

Code Breakdown:

```
#include <Servo.h>
```

```
Servo doorServo;
```

```
int irPin = 2;
```

```
int ledPin = 7;
```

```
int count = 0;
```

```
bool doorOpen = false;
```

```
void setup() {
```

```
  pinMode(irPin, INPUT);
```

```
  pinMode(ledPin, OUTPUT);
```

```
  doorServo.attach(6);
```

```
  doorServo.write(0);
```

```
}
```

```
void loop() {
```

```
  if (digitalRead(irPin) == LOW) {
```

```
    count++;
```

```
    delay(400);
```

```
  }
```

```
  if (count >= 2 && !doorOpen) {
```

```
    doorServo.write(90);
```

```
    digitalWrite(ledPin, HIGH);
```

```
    doorOpen = true;
```

```
    delay(5000);
```

```
    doorServo.write(0);
```

```
    digitalWrite(ledPin, LOW);
```

```
    count = 0;
```

```
    doorOpen = false;  
  }  
}
```

10.2 Explanation of Code

1. Initialization of Components

- The IR sensor is configured as an input device to detect hand movement.
- The LED and servo motor are set as output devices.
- The servo motor is initially set to the closed position (0°).
- A counter variable is initialized to count detections.
- A state variable is used to track whether the door is open or closed.

2. Continuous Loop Execution

- The program runs continuously in a loop.
- It repeatedly checks the IR sensor and updates the system based on input.

3. Detection of Hand Movement

- When a hand comes near the IR sensor, the output signal changes.
- The program identifies this as a valid detection.

4. Counting Mechanism

- Each valid detection increases the counter value.
- The system waits until two detections occur before taking action.

5. Debounce Logic

- A small delay is added after each detection.
- This prevents multiple counts from a single hand movement.
- It ensures stable and accurate detection.

6. Condition Checking

- The system checks if the counter is greater than or equal to two.
- It also verifies that the door is not already open.
- Only if both conditions are satisfied, the system proceeds further.

7. Door Opening Action

- The servo motor rotates to 90°, opening the door.
- The LED turns ON to indicate that the door is open.
- The system updates its state to “door open.”

8. Maintaining Open State

- The door remains open for 5 seconds.
- This allows the user to pass through comfortably.

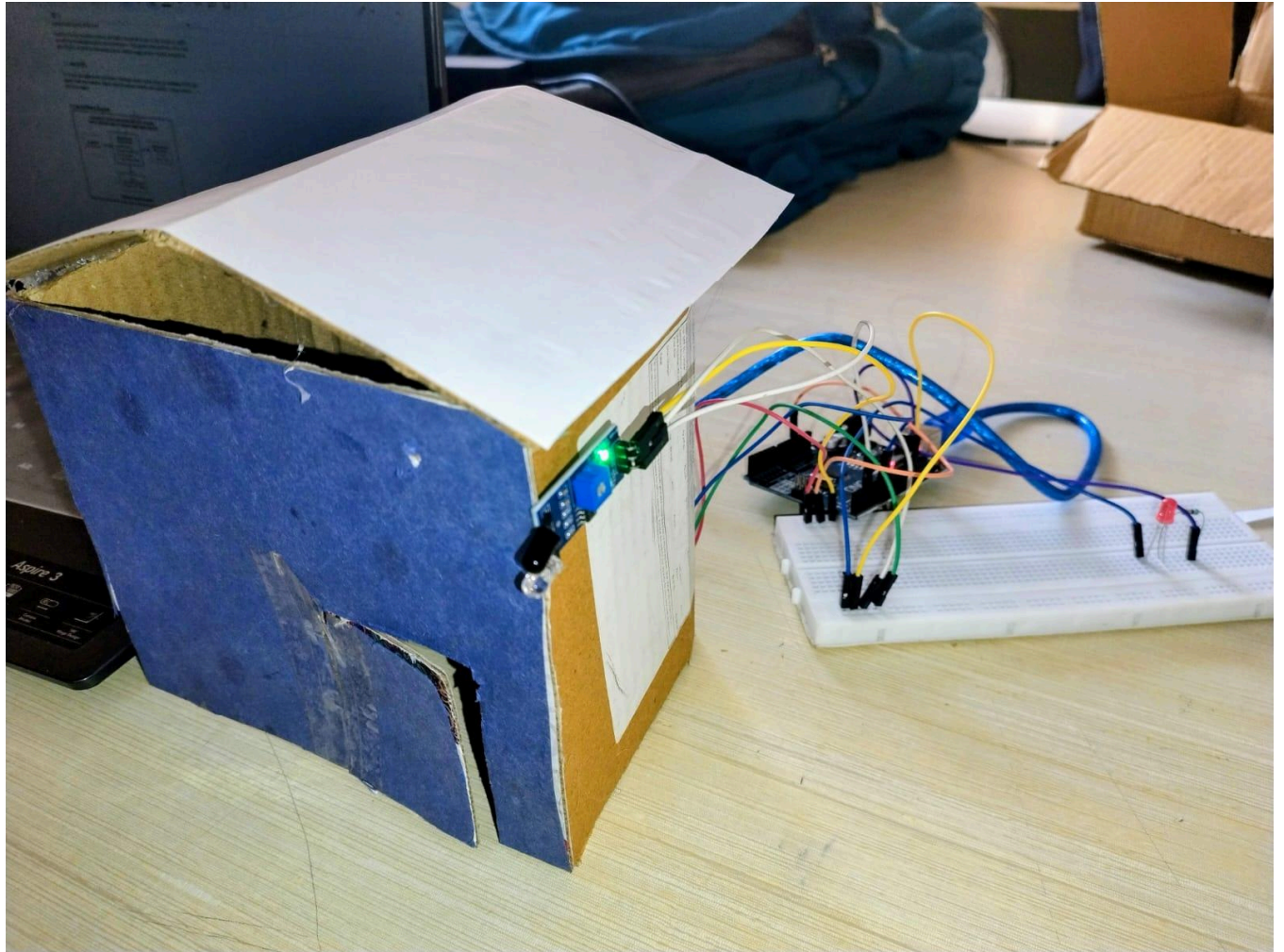
9. Door Closing Action

- After 5 seconds, the servo motor returns to 0°, closing the door.
- The LED turns OFF, indicating the door is closed.

10. System Reset

- The counter is reset to zero.
- The door state is updated to “closed.”
- The system becomes ready for the next operation cycle.

11. Photo



12. Video Link

 SA_PROJECT.mp4